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Gas detection apparatus

Abstract

A gas detection apparatus includes a light emitting part; a light receiving part; and a light guiding part for guiding light from the light emitting part to the light receiving part. When a region inside the ellipsoid E.sub.i and not including the ellipsoid E.sub.si is defined as a region R.sub.i, the ellipsoid E.sub.i including a light source region of the light emitting part is defined as an ellipsoid E.sub.s, the ellipsoid E.sub.i including a light receiving region of the light receiving part is defined as an ellipsoid E.sub.d, the region R.sub.i of the ellipsoid E.sub.s is defined as a region R.sub.in, and the region R.sub.i of the ellipsoid E.sub.d is defined as a region R.sub.out, the point G.sub.in is present in the region R.sub.in and the point Gout is present in the region R.sub.out.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2003/0136911	12/2002	Martin	N/A	N/A
2006/0086903	12/2005	Hopkins et al.	N/A	N/A
2008/0252892	12/2007	Pralle et al.	N/A	N/A
2015/0219491	12/2014	Lee et al.	N/A	N/A
2016/0231244	12/2015	Camargo et al.	N/A	N/A
2016/0252451	12/2015	Kawate	N/A	N/A
2018/0348121	12/2017	Deliwala	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
105593666	12/2015	CN	N/A
H07159382	12/1994	JP	N/A
H09184803	12/1996	JP	N/A
2003532070	12/2002	JP	N/A
2005195477	12/2004	JP	N/A
2005195685	12/2004	JP	N/A
2013002966	12/2012	JP	N/A
2014238307	12/2013	JP	N/A
2017015567	12/2016	JP	N/A
2019020305	12/2018	JP	N/A
2019174354	12/2018	JP	N/A
2019185944	12/2018	JP	N/A
1020080076515	12/2007	KR	N/A
2015056628	12/2014	WO	N/A
2015151502	12/2014	WO	N/A

OTHER PUBLICATIONS

H. C. Manoharan et al., Quantum mirages formed by coherent projection of electronic structure, Nature, Feb. 3, 2000, pp. 512-515, vol. 403. cited by applicant

Liu Xiu-Zhen et al., Study on Infrared Methane Sensor with Three Ellipsoid Absorption Cavity, Coal Technology, Oct. 2015, pp. 264-266, vol. 34, No. 10. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application is a continuation application of U.S. patent application Ser. No. 17/452,617 filed Oct. 28, 2021, which claims priority to and the benefit of Japanese Patent Application No. 2020-180845 filed Oct. 28, 2020 and Japanese Patent Application No. 2021-122620 filed Jul. 27, 2021. The disclosures of the prior applications are hereby incorporated by reference herein in their entirety.

TECHNICAL FIELD

(1) The present disclosure relates to a gas detection apparatus.

BACKGROUND

(2) Gas detection apparatuses for detecting gases have been used in various fields. For example, Patent Literature 1 (PTL 1) discloses an apparatus including a light source for emitting infrared light and a detector for detecting infrared light at a specific wavelength accommodated in a case having an ellipsoidal inner surface (ellipsoidal mirror). A gas to be detected is introduced to the case.

CITATION LIST

Patent Literature

PTL 1: US20180348121 (A1)

SUMMARY

(3) FIGS. 1A, 1B, 2A, and 2B are examples of ray tracing in an optical path design in which the centers of a light emitting part and a light receiving part are disposed at the focal positions of an ellipsoidal mirror, as disclosed in PTL 1. As illustrated in FIGS. 1A and 1B, when the ellipsoidal mirror is large enough for the size of the light emitting part, the light rays emitted from the light emitting part placed at the focal position can be collected at the light receiving part placed at the other focal point. That is, the light emitting part can be approximately regarded as a point light source. Thus, according to the general nature of ellipsoids, the light rays emitted from one focal position will focus on the other focal position.

(4) On the other hand, as illustrated in FIGS. 2A and 2B, when the ellipsoidal mirror is not large enough for the size of the light emitting part, the light rays emitted from the light emitting part are scattered throughout the ellipsoidal mirror, and cannot be collected by the light receiving part of a light receiving element. This is because a physical picture in which the light emitting part approximately behaves as a point light source is broken, and the aberration of the ellipsoidal mirror due to the size of the light emitting part will strongly affect.

(5) With the miniaturization trend of gas detection apparatus in recent years, the ratio of the size of the light emitting part to the size of the ellipsoidal mirror is decreasing.

(6) In light of the aforementioned issue, it would be thus helpful to provide a compact and highly accurate gas detection apparatus using an ellipsoidal mirror.

(7) A gas detection apparatus according to an embodiment comprises: a light emitting part; a light receiving part; and a light guiding part for guiding light from the light emitting part to the light receiving part, wherein a shape of at least a part of an inner surface of the light guiding part is

composed of all or a part of figures of n (n : one or more natural numbers) ellipsoids, and when: the n ellipsoids are defined as ellipsoids $E_{sub.1}$, $E_{sub.2}$, $\dots$, $E_{sub.(n-1)}$ and $E_{sub.n}$; an ellipse having a maximum area in a cross-section of an ellipsoid $E_{sub.i}$ (i : a natural number satisfying $1 \leq i \leq n$) is defined as an ellipse $Ec_{sub.i}$; an ellipsoid that passes through two focal points $F_{sub.ai}$ and $F_{sub.bi}$ of the ellipse $Ec_{sub.i}$ and has a minimum volume having a scaling relationship with the ellipsoid $E_{sub.i}$ without being rotated is defined as an ellipsoid $E_{sub.s}$; a region inside the ellipsoid $E_{sub.i}$ and not including the ellipsoid $E_{sub.si}$ is defined as a region $R_{sub.i}$; the ellipsoid $E_{sub.i}$ including a light source region of the light emitting part is defined as an ellipsoid $E_{sub.s}$; the ellipsoid $E_{sub.i}$ including a light receiving region of the light receiving part is defined as an ellipsoid $E_{sub.d}$; the region $R_{sub.i}$ of the ellipsoid $E_{sub.s}$ is defined as a region $R_{sub.in}$; and the region $R_{sub.i}$ of the ellipsoid $E_{sub.d}$ is defined as a region $R_{sub.out}$, 60% or more of the light source region is present in the region $R_{sub.in}$ and 60% or more of the light receiving region is present in the region $R_{sub.out}$.

(8) A gas detection apparatus according to an embodiment comprises: a light emitting part; a light receiving part; and a light guiding part for guiding light from the light emitting part to the light receiving part, wherein a shape of at least a part of an inner surface of the light guiding part is composed of all or a part of figures of n (n : one or more natural numbers) ellipsoids, and when: the n ellipsoids are defined as ellipsoids $E_{sub.1}$, $E_{sub.2}$, $\dots$, $E_{sub.(n-1)}$ and $E_{sub.n}$; an ellipse having a maximum area in a cross-section of an ellipsoid $E_{sub.i}$ (i : a natural number satisfying $1 \leq i \leq n$) is defined as an ellipse $Ec_{sub.i}$; an ellipsoid that passes through two focal points $F_{sub.ai}$ and $F_{sub.bi}$ of the ellipse $Ec_{sub.i}$ and has a minimum volume having a scaling relationship with the ellipsoid $E_{sub.1}$ without being rotated is defined as an ellipsoid $E_{sub.si}$; a region inside the ellipsoid $E_{sub.i}$ and not including the ellipsoid $E_{sub.si}$ is defined as a region $R_{sub.i}$; the ellipsoid $E_{sub.i}$ including a light source region of the light emitting part is defined as an ellipsoid $E_{sub.s}$; the ellipsoid $E_{sub.i}$ including a light receiving region of the light receiving part is defined as an ellipsoid $E_{sub.d}$; the region $R_{sub.i}$ of the ellipsoid $E_{sub.s}$ is defined as a region $R_{sub.in}$; the region $R_{sub.i}$ of the ellipsoid $E_{sub.d}$ is defined as a region $R_{sub.out}$; the center of gravity of the light source region or a peak point of luminance is defined as a point $G_{sub.in}$; and the center of gravity of the light receiving region is defined as a point G_{out} , the point $G_{sub.in}$ is present in the region $R_{sub.in}$ and the point G_{out} is present in the region $R_{sub.out}$.

(9) According to an embodiment of the present disclosure, a compact and highly accurate gas detection apparatus using an ellipsoidal mirror can be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In the accompanying drawings:
- (2) FIGS. 1A-1B are diagrams illustrating an example of ray tracing in an ellipsoidal mirror;
- (3) FIGS. 2A-2B are diagrams illustrating another example of ray tracing in an ellipsoidal mirror;
- (4) FIG. 3 is a partially transparent perspective view of a gas detection apparatus according to an embodiment;
- (5) FIGS. 4A-4B are diagrams illustrating a ray tracing simulation result in a spheroidal mirror;
- (6) FIG. 5 is a diagram for explaining a region $R_{sub.i}$;
- (7) FIGS. 6A-6B are diagrams illustrating a configuration example of the gas detection apparatus according to the present embodiment;
- (8) FIG. 7 is a variation of the gas detection apparatus according to the present embodiment;
- (9) FIG. 8 is a diagram illustrating a ray simulation result of a general ellipsoid;
- (10) FIG. 9 is a diagram illustrating a configuration example of a light emitting part including a passive element and a light receiving part including an indirect element; and

(11) FIG. 10 is a diagram illustrating another configuration example of a light emitting part including a passive element and a light receiving part including an indirect element.

DETAILED DESCRIPTION

(12) <Gas Detection Apparatus of the Present Embodiment>

(13) A gas detection apparatus of the present embodiment comprises a light emitting part, a light receiving part, and a light guiding part for guiding light from the light emitting part to the light receiving part.

(14) The shape of at least a part of an inner surface of the light guiding part is composed of all or a part of figures of n (n : one or more natural numbers) spheroids.

(15) The n spheroids are defined as ellipsoids $E_{\text{sub}.1}$, $E_{\text{sub}.2}$, . . . , $E_{\text{sub}.(n-1)}$ and $E_{\text{sub}.n}$. An ellipsoid that passes through two focal points $F_{\text{sub}.ai}$ and $F_{\text{sub}.bi}$ of an ellipsoid $E_{\text{sub}.i}$ (i : a natural number satisfying $1 \leq i \leq n$), has the same axis of rotational symmetry as that of the ellipsoid $E_{\text{sub}.i}$, and has a similarity shrinking relationship with the ellipsoid $E_{\text{sub}.i}$ is defined as an ellipsoid $E_{\text{sub}.si}$. A region inside the ellipsoid $E_{\text{sub}.i}$ and not including the ellipsoid $E_{\text{sub}.si}$ is defined as a region $R_{\text{sub}.i}$. The ellipsoid $E_{\text{sub}.i}$ including a light source region of the light emitting part is defined as an ellipsoid $E_{\text{sub}.s}$. The ellipsoid $E_{\text{sub}.i}$ including a light receiving region of the light receiving part is defined as an ellipsoid $E_{\text{sub}.d}$ (when the number of ellipsoids is one, that is, when $n=1$, $E_{\text{sub}.s}=E_{\text{sub}.d}$). A region $R_{\text{sub}.i}$ of the ellipsoid $E_{\text{sub}.s}$ is defined as a region $R_{\text{sub}.in}$. A region $R_{\text{sub}.i}$ of the ellipsoid $E_{\text{sub}.d}$ is defined as a region $R_{\text{sub}.out}$.

(16) In the gas detection apparatus according to the present embodiment, 60% or more of an area of the light source region is present in the region $R_{\text{sub}.in}$ and 60% or more of an area of the light receiving region is present in the region $R_{\text{sub}.out}$.

(17) Although the detailed principle will be described later, with this configuration, a compact and highly accurate gas detection apparatus using an ellipsoidal mirror can be provided.

(18) <Example of Specific Configuration of Gas Detection Apparatus of the Present Embodiment>

(19) FIG. 3 is a partially transparent perspective view of a gas detection apparatus according to an embodiment of the present disclosure. As an example, the gas detection apparatus is a compact apparatus with length, width and height of 7 mm, 5 mm and 3 mm, respectively, and is also referred to as a gas sensor.

(20) In the present embodiment, the gas detection apparatus is a Non Dispersive InfraRed (NDIR) type apparatus that measures the concentration of the gas to be detected based on the infrared rays transmitted through the introduced gas.

(21) The gas to be detected is, for example, carbon dioxide, water vapor, carbon monoxide, nitric oxide, ammonia, sulfur dioxide or alcohol, hydrocarbon gases such as formaldehyde, methane, and propane.

(22) <Architecture (Interrelationship of Components)>

(23) The gas detection apparatus comprises a light emitting part, a light receiving part and a light guiding part. The gas detection apparatus may further comprise a holding part. Further, the gas detection apparatus may comprise an additional controller. The gas detection apparatus according to an embodiment of the present disclosure illustrated in FIG. 3 comprises a light receiving part **20** and a light emitting part **10** held by a holding part **40** and a light guiding part **30** held by the holding part **40**. Although not illustrated, a controller for controlling at least one of the light emitting part **10** and the light receiving part **20** may be provided in the holding part **40**.

(24) The surfaces of the light emitting part **10** and the light receiving part **20** are facing to the space (detection space) between the inner wall of the light guiding part **30** and the upper surface of the holding part **40**. Further, the light guiding part **30** has a gas port **31** that can introduce and withdraw gas into and from the detection space.

(25) The light emitted from the light emitting part **10** is reflected at least once on the inner surface of the light guiding part **30** and reaches the light receiving part **20**.

(26) Next, a detailed description of the components of the gas detection apparatus according to the

present embodiment will be given.

(27) <Light Emitting Part>

(28) The light emitting part **10** is a component that emits light used for detection of the gas to be detected. The light emitting part **10** is not particularly limited as long as it outputs light that includes a wavelength absorbed by the gas to be detected. In the present embodiment, the light emitted by the light emitting part **10** is infrared rays, but it is not limited thereto.

(29) The light emitting part **10** has a light emitting element **10A**. In the present embodiment, although the light emitting element **10A** is a light emitting diode (LED), as the other examples, it can be a lamp, a laser (Light Amplification by Stimulated Emission of Radiation), an organic light emitting element or Micro Electro Mechanical Systems (MEMS) heater, etc. Further, the light emitting part **10** may include not only the light emitting element **10A** but also a passive element that passively emits light by receiving the light emitted by the light emitting element **10A**.

Specifically, the passive element is a mirror, an optical filter, a phosphor, an optical image, an optical fiber, an optical waveguide, a lens, or a diffraction grating.

(30) In this embodiment, since the light emitting part **10** has only the light emitting element **10A**, the light emitting part **10** and the light emitting element **10A** refer to the same thing.

(31) The light emitting part **10** has a light source region. The light source region is a region in which photons of the light emitting element **10A** are generated when the light is directly guided from the light emitting element **10A** to the light guiding part **30** without passing through the passive element as in the present embodiment. Specifically, for a quantum type light emitting element **10A**, it is an active region, and for a thermal light source, it is a high temperature region. For example, in the case of a lamp, it is a filament.

(32) Further, when the light emitting part **10** includes a passive element and the light emitted by the light emitting element **10A** is guided to the light guiding part **30** through the passive element, the light source region is an aggregate of the light emitting ends of the passive element. Specifically, when the passive element is a mirror, the light source region is a region that reflects light rays.

(33) Further, when the passive element is an optical filter having a wavelength selection function, the region through which the light rays pass on the surface where the optical filter is in contact with the space may be regarded as the light source region. Further, when the passive element is an optical fiber, an optical waveguide, or a lens, the emission surface through which the light rays pass on the surface in contact with the space may be regarded as the light source region.

(34) Further, when an optical image is formed as the light emitting part **10** by a lens, a mirror, or the like, the formed image may be regarded as a light source region.

(35) Here, the light emitting element **10A** is preferably a planar surface light source such as an LED, a MEMS heater, or a Vertical Cavity Surface Emitting Laser (VCSEL).

(36) <Light Receiving Part>

(37) The light receiving part **20** is a component that receives light transmitted through the introduced gas. The light receiving part **20** is not particularly limited as long as it has sensitivity in the bandwidth of light including the wavelength absorbed by the gas to be detected. In the present embodiment, the light received by the light receiving part **20** is infrared rays, but the light is not limited to this.

(38) The light receiving part **20** has a light receiving element **20A**. In the present embodiment, the light receiving element **20A** is a photodiode, but the other examples may include a phototransistor, a thermopile, a pyroelectric sensor, a bolometer, a photoacoustic detector, or the like. Further, the light receiving part **20** may include not only the light receiving element **20A** but also an indirect element that guides light to the light receiving element **20A**. Specifically, the indirect elements are mirrors, optical filters, phosphors, lenses, diffraction gratings, optical fibers, and optical waveguides.

(39) In the present embodiment, since the light receiving part **20** has only the light receiving element **20A**, the light receiving part **20** and the light receiving element **20A** refer to the same

thing.

(40) The light receiving part **20** has a light receiving region. The light receiving region is a region in the light receiving element **20A** having a function of converting the received light into a signal when the light receiving element **20A** directly receives light without passing through an indirect element as in the present embodiment. Specifically, when the light receiving element **20A** is a photodiode, the light receiving region is an active layer, and when the light receiving element **20A** is a thermopile, it is a thermoelectric converting part.

(41) Further, in the light receiving part **20**, when the light receiving element **20A** receives light through the indirect element, the light receiving region is a region, in the indirect element, that has an optical function for guiding the received light to the light receiving element **20A**, and through which light rays pass. Specifically, when the indirect element is an optical filter having a wavelength selection function, the region through which the light rays pass on the surface of the optical filter in contact with the space may be regarded as the light receiving region. Further, when the indirect element is an optical fiber, an optical waveguide, or a lens, the incident surface through which the light rays pass on the surface in contact with the space may be regarded as the light receiving region. When the indirect element is a mirror, the light receiving region is a region that reflects light rays.

(42) <Light Guiding Part>

(43) The light guiding part **30** is a member that guides the light emitted by the light emitting part **10** to the light receiving part **20**, and is an optical system of the gas detection apparatus. The light emitted from the light emitting part **10** is reflected by the light guiding part **30** and reaches the light receiving part **20**. In other words, the light guiding part **30** optically connects the light emitting part **10** and the light receiving part **20**.

(44) In the present embodiment, the inner surface of the light guiding part **30** is a mirror (reflection surface). The shape of at least a part of the inner surface is a figure of all or a part of a spheroid. The light guiding part **30** may further comprise a flat mirror, a concave mirror or a convex mirror, a lens, and a diffraction grating as an auxiliary.

(45) The material constituting the mirror may be, for example, metal, glass, ceramics, stainless steel, etc., but is not limited to them.

(46) From the viewpoint of improving the detection sensitivity, these mirrors are preferably made of materials having a low light absorption coefficient and a high reflectance. Specifically, resin housings provided with coating of an alloy containing aluminum, gold, or silver, a dielectric, or a laminate of these materials are preferred. Examples of the materials of the resin housing include liquid crystal polymer (LCP), polypropylene (PP), polyether ether ketone (PEEK), polyamide (PA), polyphenylene ether (PPE), polycarbonate (PC), polyphenylene sulfide (PPS), polymethyl methacrylate resin (PMMA), polyarylate resin (PAR), or the like, and hard resins of a mixture of two or more of these. Further, resin housings coated with gold or an alloy layer containing gold are preferred in view of the reliability and degradation over time. Moreover, a laminated film of a dielectric is preferably formed on the surface of the metal layer for increasing the reflectance. Formation of the inner surface of the light guiding part **30** on the resin housing by vapor deposition or plating can achieve a higher productivity and provide improved weight reduction as compared with cases where they are made from a metal material. Furthermore, the difference in the thermal expansion coefficients with the holding part **40** is reduced, which suppresses thermal deformations and results in less fluctuation of sensitivity.

(47) Further, the light guiding part **30** may be formed by cutting machining, and is preferably formed by injection molding in view of the productivity.

(48) <Holding Part>

(49) The holding part **40** is a member for holding the light receiving part **20**, the light emitting part **10** and the light guiding part **30**. Holding means trying to maintain the relative positional relationship of each member with respect to an external force. The holding form is not particularly

limited, but mechanical holding is preferable. The holding form may be electromagnetic or chemical holding.

(50) When the gas detection apparatus of the present embodiment has a controller, the controller may be held by the holding part **40**.

(51) The holding part **40** is not particularly limited as long as it can hold the light receiving part **20**, the light emitting part **10**, and the light guiding part **30**. In the present embodiment, the holding part **40** is a resin package, but as the other examples, it may be a printed circuit board or a ceramic package. Alternatively, a semiconductor substrate may be used as the holding part **40**, and the light receiving part **20** and the light emitting part **10** may be formed on the same semiconductor substrate. When the holding part **40** is a resin package, it may have a lead frame inside thereof, and the lead frame, the light emitting part **10**, the light receiving part **20**, and the controller may be electrically connected by a wire or the like. Further, when the holding part **40** is a printed circuit board, the printed circuit board, the light receiving part **20**, and the light emitting part **10** may be electrically and mechanically connected by solder. Moreover, the holding part **40** and the light guiding part **30** are mechanically held by an adhesive, screws, claws, fittings, grommets, welding and the like. Further, the holding part **40** may have a connection terminal for making an electrical connection with the outside.

(52) <Controller>

(53) The controller is a member that controls at least one of the light emitting part **10** and the light receiving part **20**. The controller may have an analog-to-digital conversion circuit that converts an analog electric signal output from the light receiving part **20** into a digital electric signal. Moreover, the controller may have an operation part that executes a gas concentration calculation based on the converted digital electric signal.

(54) The controller may have at least one of a general-purpose processor that executes a function corresponding to a read-in program and a dedicated processor for a specific process. The dedicated processor may include an Application Specific Integrated Circuit (ASIC). The processor may include a Programmable Logic Device (PLD).

(55) <Size of Gas Detection Apparatus>

(56) FIGS. **1A** and **1B** illustrate that when the ellipsoidal mirror is sufficiently large with respect to the size of the light emitting part **10**, the light rays emitted from the light emitting part **10** can be collected by the light receiving part **20**. Here, as the size of the light guiding part **30**, a part of the shape of the light guiding part **30** is configured as an ellipsoid, of which a maximum length of the ellipsoid including the light emitting part **10** is defined as L_{ms} and a maximum length of the light source region is defined as L_s . Here, when the condition of $L_s < L_{ms}/50$ is satisfied, the size of the light emitting part **10** is sufficiently small with respect to the ellipsoidal mirror, and it can be approximately regarded as a point light source. Thus, the light rays emitted from one focal point position are collected at the other focal point position. On the other hand, as illustrated in FIGS. **2A** and **2B**, when the ellipsoidal mirror is not sufficiently large with respect to the size of the light emitting part **10** ($L_s \geq L_{ms}/50$), the light rays emitted from the light emitting part **10** are scattered throughout the entire ellipsoidal mirror, and cannot be collected at the light receiving part **20**.

(57) Although not particularly limited, the gas detection apparatus of the present embodiment exerts a particularly remarkable effect when $L_s \geq L_{ms}/50$.

(58) Although not particularly limited, similarly, the gas detection apparatus of the present embodiment exerts a particularly remarkable effect when $L_d \geq L_{md}/50$, where a maximum length of the ellipsoid including the light receiving part **20** is L_{md} and a maximum length of the light receiving part **20** is L_d .

(59) Next, the principle of the gas detection apparatus of the present embodiment will be described in detail with reference to the drawings.

(60) <Explanation of Principle>

(61) FIGS. **4A** and **4B** are diagrams for explaining the basic principle of the present embodiment,

and is a ray tracing simulation result in a plane passing through the rotation axis of a spheroidal mirror. As described above, the gas detection apparatus may also be composed of a plurality of spheroids, but one spheroid will be used for explaining the basic principle. When n (n : one or more natural numbers) ellipsoids constituting the gas detection apparatus are defined as ellipsoids $E_{\text{sub.1}}, \dots, E_{\text{sub.i}}, \dots, \text{and } E_{\text{sub.n}}$ (i : a natural number satisfying $1 \leq i \leq n$), the spheroid in each of FIGS. 4A, 4B and 5 corresponds to one ellipsoid $E_{\text{sub.i}}$ ($i=n=1$).

(62) FIG. 4A illustrates a light ray simulation result when light rays are emitted from a point of the region $R_{\text{sub.i}}$, which is a region outside the focal points $F_{\text{sub.ai}}$ and $F_{\text{sub.bi}}$ of the spheroid. At this time, the light rays are repeatedly reflected on the mirror surface. However, the light rays do not enter the region $R_{\text{sub.INTER}}$, which is a region that connects between focal points near the center of ellipsoid.

(63) Further, FIG. 4B is a light ray simulation result when the light rays are emitted from the point of the region $R_{\text{sub.INTER}}$. The light rays do not enter the region $R_{\text{sub.EDGE}}$, which is the region at both ends outside the focal points $F_{\text{sub.ai}}$ and $F_{\text{sub.bi}}$ of the spheroid.

(64) The separation phenomenon of the light ray penetration region in this region $R_{\text{sub.i}}$ and the region $R_{\text{sub.INTER}}$ (hereinafter referred to as a region separation phenomenon) is explained by equating the light ray trajectory with the trajectory of a rigid sphere that repeatedly and elastically collides with a wall having the same shape as that of the mirror surface. The general angular momentum $J = L_{\text{sub.Fai}} \cdot \sin \theta_{\text{Fbi}}$, which is an inner product of the angular momentum $L_{\text{sub.Fai}}$ centered on the focal point $F_{\text{sub.ai}}$ and the angular momentum $L_{\text{sub.Fbi}}$ centered on the focal point $F_{\text{sub.bi}}$ is conserved, with respect to the movement of a rigid sphere that elastically reflects on an ellipsoidal wall in free space. At this time, since the rotation directions viewed from each focal point are the same in the trajectory of the rigid sphere emitted from the point in the region $R_{\text{sub.i}}$, the angular momentums $L_{\text{sub.Fai}}$ and $L_{\text{sub.Fbi}}$ are in the same direction, and the general angular momentum J is positive ($J > 0$). On the other hand, when the trajectory passes through the region $R_{\text{sub.INTER}}$, since the rotation directions viewed from each focal point are opposite, the general angular momentum J is negative ($J < 0$).

(65) That is, according to the conservation law of general angular momentum J , a trajectory (light ray) of a rigid sphere emitted first from the region $R_{\text{sub.i}}$ has a positive value of general angular momentum J , and it cannot be a trajectory (light ray) of a rigid sphere passing through the region $R_{\text{sub.INTER}}$ with the general angular momentum of a negative value, no matter how many times the trajectory is reflected by the wall on the mirror. On the contrary, a trajectory (light ray) of a rigid sphere emitted from the region $R_{\text{sub.INTER}}$ with the general angular momentum of a negative value cannot be a trajectory (light ray) of a rigid sphere passing through the region $R_{\text{sub.i}}$ with the general angular momentum of a positive value, no matter how many times the trajectory is reflected by the wall on the mirror. In this manner, the region separation phenomenon occurs. Here, when the general angular momentum J is 0, it corresponds to a trajectory emitted from one focal point and reaches the other focal point, whereby the phase space of the rigid sphere motion is separated.

(66) Here, when an ellipse having a maximum area in the cross section of the ellipsoid $E_{\text{sub.i}}$ (i : a natural number satisfying $1 \leq i \leq n$) is defined as an ellipse $E_{\text{sub.ci}}$, the focal point is uniquely determined if it is a spheroid, and the maximum area is a cross section passing through two focal points. Further, when an ellipsoid that passes through the two focal points $F_{\text{sub.ai}}$ and $F_{\text{sub.bi}}$ of the ellipsoid $E_{\text{sub.ci}}$ and has a minimum volume having a scaling relationship with the ellipsoid $E_{\text{sub.i}}$ without being rotated is defined as an ellipsoid $E_{\text{sub.si}}$, the focal point is uniquely determined if it is a spheroid. Thus, the above described principle generally holds “when an ellipsoid having a maximum area in the cross section of the ellipsoid $E_{\text{sub.i}}$ (i : a natural number satisfying $1 \leq i \leq n$) is defined as an ellipsoid $E_{\text{sub.ci}}$ and an ellipsoid that passes through the two focal points $F_{\text{sub.ai}}$ and $F_{\text{sub.bi}}$ of the ellipsoid $E_{\text{sub.ci}}$ and has a minimum volume having a scaling relationship with the ellipsoid $E_{\text{sub.i}}$ without being rotated is defined as an ellipsoid

E.sub.si.”

(67) Next, the configuration of the gas detection apparatus of the present embodiment based on the above-described separation phenomenon of the light ray penetrating region will be described.

(68) FIG. 5 is a diagram for explaining the present embodiment that applies the region separation phenomenon to the inner surface of the light guiding part **30** that includes one spheroidal shape. Due to the region separation phenomenon, when at least a part of the light source region is disposed in the region R.sub.i, the light rays emitted from the light source at this position are always present in the region R.sub.i even if they are repeatedly reflected by the mirror. At this time, when the light receiving region is also disposed in the region R.sub.i, the light rays stay in the region R.sub.i without scattering over the region R.sub.INTER, and thus light can be collected efficiently with respect to the light receiving region.

(69) In this manner, the gas sensitivity of the detection apparatus can be improved. Even in the case where the light source region and the light receiving region are present partially in the region R.sub.i, the effect of the present embodiment is exhibited in the partial region. Therefore, the effect is realized when 60% or more of the light source region is present in the region R.sub.i. Further, in the complementary same expression, the effect is realized when less than 40% of the light source region and the light receiving region are present in the region R.sub.INTER. From the viewpoint of improving gas sensitivity, preferably, 70% or more of the light source region is present in the region R.sub.i, more preferably, 80% or more of the light source region is present in the region R.sub.i, and preferably, the entire light source region is present in the region R.sub.i. Similarly, from the viewpoint of improving gas sensitivity, preferably, 70% or more of the light receiving region is present in the region R.sub.i, more preferably, 80% or more of the light receiving region is present in the region R.sub.i, and preferably, the entire light receiving region is present in the region R.sub.i.

(70) That is, when an ellipsoid in which at least a part of the shape of the inner surface of the light guiding part **30** passes through the two focal points F.sub.ai and F.sub.bi of the ellipsoid E.sub.i, and that has the same axis of rotational symmetry as that of the ellipsoid E.sub.i and a similarity shrinking relationship with the ellipsoid E.sub.i is defined as an ellipsoid E.sub.si, if 60% or more of the areas of the light source region of the light emitting part **10** and the light receiving region of the light receiving part **20** are present in the region R.sub.i, which is a region inside the ellipsoid E.sub.i and not including the ellipsoid E.sub.si, a compact and highly accurate gas detection apparatus using an ellipsoidal mirror is realized. This effect is noticeable when the two focal points of the spheroid are sufficiently separated and the region R.sub.INTER is formed. Specifically, this effect is remarkably exhibited when the ratio of the semi-minor axis b and the semi-major axis a of the spheroid, “a/b,” is 1.2 or more. “b” is the shortest semi-minor axis b.

(71) Here, although it has been described that 60% or more of the light source region is present in the region R.sub.in and 60% or more of the light receiving region is present in the region R.sub.out, focusing on the center of gravity or the peak point of luminance, the following holds. That is, assuming that the center of gravity of the light source region or the peak point of luminance is a point G.sub.in and the center of gravity of the light receiving region is a point G.sub.out, the point G.sub.in is present in the region R.sub.in and the point G.sub.out is present in the region R.sub.out.

(72) <Arrangement in a Plurality of Ellipsoids>

(73) Although an embodiment in which one spheroidal mirror (i=1) is used has been described above, the gas detection apparatus of the present embodiment exhibits the same effect even if it comprises the light guiding part **30** having a plurality of spheroidal mirrors.

(74) FIGS. 6A and 6B illustrate a gas detection apparatus according to the present embodiment that uses three light guiding parts **30** each having a spheroidal inner surface (reflective surface). FIG. 6A is a schematic diagram and FIG. 6B is a light ray tracing simulation result. The three light guiding parts **30** are composed of three ellipses and may be one connected member as a whole.

(75) In FIG. 6A, the region R.sub.1 with respect to the ellipsoid E.sub.1 containing the light source

region of the spheroid is defined as a region R.sub.in. A neighboring ellipsoid E.sub.2 includes a region R.sub.2. When the light source region is located in the region R.sub.in, the light rays are brought, without scattering, into the region R.sub.in on the other end side of the ellipsoid E.sub.s1. When the region into which the light rays are brought is regarded again as a light source of the ellipsoid E.sub.2, the light rays can move in the region R.sub.2 without scattering. In order to realize this efficiently, it is preferable that, for reflecting the light rays, auxiliary reflectors **50** as illustrated in FIGS. **6A** and **6B** are further provided as light guiding parts **30** for the neighboring spheroid. Although the form of the auxiliary reflector **50** is not particularly limited, examples thereof include a flat mirror, a concave mirror, a convex mirror, etc. The auxiliary reflector **50** may be formed of a figure different from that of the spheroid. From the viewpoint of simplicity and efficiency, the auxiliary reflector **50** is preferably a flat mirror. Further, the auxiliary reflector **50** may have a wavelength selecting function. The auxiliary reflector **50** may preferably be disposed in the region R.sub.i of a spheroid.

(76) By repeating the above sequentially, the light rays can move without scattering as illustrated in the light ray tracing simulation result of FIG. **6B**. Finally, when the light receiving region is disposed at last in the region R.sub.out, which is the region R.sub.3 of the ellipsoid E.sub.3, light can be collected efficiently in the same manner as in the case where one spheroidal mirror is used.

(77) That is, when 60% or more of the area of the light source region of the light emitting part **10** is present in the region R.sub.in and 60% or more of the area of the light receiving region of the light receiving part **20** is present in the region R.sub.out, a compact and highly accurate gas detection apparatus using an ellipsoidal mirror can be realized. It is possible to design an optical path length longer than in the case of one spheroidal mirror, which may be preferable from the viewpoint of detection accuracy.

(78) FIG. **7** is a variation where two spheroids are used. The light guiding part **30** is composed of a mirror in which two spheroids are combined and a plane mirror, and the long axes of the two spheroids intersect at right angles. At this time, when the auxiliary reflector **50** is provided in the region R.sub.i, as with the case of FIGS. **6A** and **6B**, a compact and highly accurate gas detection apparatus using an ellipsoidal mirror can be realized.

(79) As described above, although the light guiding part **30** has been described as a spheroid, it may be a general ellipse, that is, one having no rotational symmetry and having three different diameters. Since a rigid sphere system that elastically reflects inside a general ellipsoidal wall (so-called a billiard problem) is also an integrable system, there is a conserved quantity similar to the general angular momentum J , and a similar region separation phenomenon follows. FIG. **8** illustrates a light ray simulation result when the light rays are emitted from the point of the region R.sub.i, which is a region outside the focal points F.sub.ai and F.sub.bi of a general ellipsoid. The focal points F.sub.ai and F.sub.bi illustrated in FIG. **8** are focal points on a plane ellipse that has a maximum area when the ellipsoid is cut in a plane, and the light rays emitted from the region R.sub.i outside thereof continue to stay in the region R.sub.i even if repeatedly reflected. This is also understood from the fact that when the smallest one of the ellipse diameters becomes ultimately zero, it will be the same as the region separation phenomenon due to the focal point in a plane ellipse, and a general ellipse case is positioned between the spheroid case described above and this plane ellipse case.

(80) Although the embodiments have been described with reference to the drawings and the examples, it should be noted that various modifications and variations can be readily conceived of by a person skilled in the art based on the present disclosure. Therefore, it should be understood that such modifications and variations are encompassed within the scope of the present disclosure. For example, the members, the functions included in each means, or the like can be rearranged unless they are logically contradicted, and a plurality of means, and the like can be combined into one or means can be divided, for example.

(81) FIGS. **9** and **10** illustrate configuration examples of the light emitting part **10** including a

passive element and a light receiving part **20** including an indirect element according to another embodiment. In the example illustrated in FIG. **9**, the light emitting part **10** includes a light emitting element **10A** and a passive element, which is a 45° mirror. Further, in the example illustrated in FIG. **9**, the light receiving part **20** includes a light receiving element **20A** and an indirect element, which is a 45° mirror. In the example illustrated in FIG. **10**, the light emitting part **10** includes the light emitting element **10A** and a passive element, which is a lens. Further, in the example illustrated in FIG. **10**, the light receiving part **20** includes a light receiving element **20A** and an indirect element, which is a concave mirror.

(82) (Appendant)

(83) A gas detection apparatus according to an embodiment comprises: a light emitting part; a light receiving part; and a light guiding part for guiding light from the light emitting part to the light receiving part, wherein a shape of at least a part of an inner surface of the light guiding part is composed of all or a part of figures of n (n : one or more natural numbers) spheroids; and when: the n spheroids are defined as ellipsoids $E_{sub.1}$, $E_{sub.2}$, . . . , $E_{sub.(n-1)}$ and $E_{sub.n}$; an ellipse that passes through two focal points $F_{sub.ai}$ and $F_{sub.bi}$ of an ellipsoid $E_{sub.i}$ (i : a natural number satisfying $1 \leq i \leq n$), has the same axis of rotational symmetry as that of the ellipsoid $E_{sub.i}$, and has a similarity shrinking relationship with the ellipsoid $E_{sub.i}$ is defined as an ellipsoid $E_{sub.si}$; a region inside the ellipsoid $E_{sub.i}$ and not including the ellipsoid $E_{sub.si}$ is defined as a region $R_{sub.i}$; the ellipsoid $E_{sub.i}$ including a light emitting surface of the light emitting part is defined as an ellipsoid $E_{sub.s}$; the ellipsoid $E_{sub.i}$ including a light receiving surface of the light receiving part is defined as an ellipsoid $E_{sub.d}$; the region $R_{sub.i}$ of the ellipsoid $E_{sub.s}$ is defined as a region $R_{sub.in}$, and the region $R_{sub.i}$ of the ellipsoid $E_{sub.d}$ is defined as a region $R_{sub.out}$, 60% or more of an area of the light emitting surface may be present in the region $R_{sub.in}$ and 60% or more of an area of the light receiving surface may be present in the region $R_{sub.out}$.

Claims

1. A gas detection apparatus comprising: a light emitting part; a light receiving part; and a light guiding part for guiding light from the light emitting part to the light receiving part, wherein a shape of at least a part of an inner surface of the light guiding part is composed of all or a part of n (n : one or more natural numbers) ellipsoids, and when: the n ellipsoids are defined as ellipsoids $E_{sub.1}$, $E_{sub.2}$, . . . , $E_{sub.(n-1)}$ and $E_{sub.n}$; an ellipse having a maximum area in a cross-section of an ellipsoid $E_{sub.i}$ (i : a natural number satisfying $1 \leq i \leq n$) is defined as an ellipse $E_{c.sub.i}$; an ellipsoid that passes through two focal points $F_{sub.ai}$ and $F_{sub.bi}$ of the ellipse $E_{c.sub.i}$ and has a minimum volume having a scaling relationship with the ellipsoid $E_{sub.i}$; without being rotated is defined as an ellipsoid $E_{sub.si}$; a region inside the ellipsoid $E_{sub.i}$ and not including the ellipsoid $E_{sub.si}$ is defined as a region $R_{sub.i}$; the ellipsoid $E_{sub.i}$ including a light source region of the light emitting part is defined as an ellipsoid $E_{sub.s}$; the ellipsoid $E_{sub.i}$ including a light receiving region of the light receiving part is defined as an ellipsoid $E_{sub.d}$; the region $R_{sub.i}$ of the ellipsoid $E_{sub.s}$ is defined as a region $R_{sub.in}$; and the region $R_{sub.i}$ of the ellipsoid $E_{sub.d}$ is defined as a region $R_{sub.out}$; the center of gravity of the light source region or a peak point of luminance is defined as a point $G_{sub.in}$; and the center of gravity of the light receiving region is defined as a point $G_{sub.out}$, the point $G_{sub.in}$ is present in the region $R_{sub.in}$ and the point $G_{sub.out}$ is present in the region $R_{sub.out}$, wherein the inner surface of the light guiding part is reflective; wherein, when a maximum length of the light source region is defined as $L_{sub.s}$ and a maximum length of the ellipsoid $E_{sub.s}$ is defined as $L_{sub.ms}$, $L_{sub.s} \geq L_{sub.ms}/50$; and wherein, when a maximum length of the light receiving part is defined as $L_{sub.d}$ and a maximum length of the ellipsoid $E_{sub.d}$ is defined as $L_{sub.md}$, $L_{sub.d} \geq L_{sub.md}/50$.

2. The gas detection apparatus according to claim 1, wherein a ratio a/b of a semi-major axis a and a semi-minor axis b of each of the ellipsoid $E_{sub.s}$ and the ellipsoid $E_{sub.d}$ is 1.2 or more.

3. The gas detection apparatus according to claim 1, wherein an identical holding part holds the light emitting part and the light receiving part.
 4. The gas detection apparatus according to claim 3, wherein the identical holding part further holds a controller.
 5. The gas detection apparatus according to claim 1, wherein at least one of the light receiving part and the light emitting part has an optical filter.
 6. The gas detection apparatus according to claim 1, further comprising an auxiliary reflector composed of a figure different from the n ellipsoids.
 7. The gas detection apparatus according to claim 6, wherein the auxiliary reflector is present in the region R.sub.i.
 8. The gas detection apparatus according to claim 1, wherein the light emitting part is a plane light source.
 9. The gas detection apparatus according to claim 1, wherein the n ellipsoids are spheroids.
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